



Qualification Pack



Service Assistant (Agriculture Machineries)

QP Code: WBSC/AGR/Q3201

Version: 1.0

NSQF Level: 3

WB State Council of Technical and Vocational Education and Skill Dev || 5th Floor, Karigori Bhawan,
Plot- b/7, Action Area-III, New Town
Rajarhat || email:pinaki@wbscte.ac.in



Qualification Pack

Contents

WBSC/AGR/Q3201: Service Assistant (Agriculture Machineries)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	4
WBSC/AGR/N3201: Apply Safe Working Practices (Service Assistant Agriculture Machineries)	6
WBSC/AGR/N3202: Identify and select various agricultural machineries and describe the uses of each.	9
WBSC/AGR/N3203: Describe the operations of Tillage implements, demonstrate its working, identify the faults and repair.	13
WBSC/AGR/N3204: Describe the operations of seeding machinery, its working, identify the faults and repair.	17
WBSC/AGR/N3205: Recognize different components, different faults, their cause & repair minor faults concern with Harvesting machineries and Threshing machineries.	21
WBSC/AGR/N3206: Classify IC engines used in agriculture, explain the working of engine system, fuel system, cooling system, lubricating system, inlet & exhaust system with the help of flow charts	25
WBSC/AGR/N3207: Identify and Describe different component and systems of Tractor and repair minor faults.	29
WBSC/AGR/N3208: Identify and Recognize different component and systems of Power Tiller and rectify minor faults	32
WBSC/AGR/N3209: Demonstrate different component and systems of Plant Protection Equipment and rectify minor faults.	35
WBSC/AGR/N3210: Exhibit different component and systems of Irrigation Equipment and rectify minor faults	38
WBSC/AGR/N3211: Demonstrate different component and systems of electric motor used in Pump Set for Agricultural use and rectify minor faults	42
WBSC/AGR/N3212: Understand principles of pump set, their components, common faults and dismantling different parts	46
WBSC/AGR/N3213: Calculate rough estimation and costing of repair work for each machinery.	49
WBSC/AGR/N3214: Able to work in real job situation with special emphasis on basic safety and hazards in this domain.	52
DGT/VSQ/N0102: Employability Skills (60 Hours)	55
Assessment Guidelines and Weightage	62
<i>Assessment Guidelines</i>	62
<i>Assessment Weightage</i>	63
Acronyms	66
Glossary	67



Qualification Pack

WBSC/AGR/Q3201: Service Assistant (Agriculture Machineries)

Brief Job Description

Perform repair and servicing of Agriculture machineries like Tillage, Seeding, Harvesting, Threshing machineries and Tiller, Plant Protection equipment along with Diesel pump set.

Personal Attributes

The individual should possess a keen sense of Agricultural process, Mechanical Aptitude, Attention to Detail, and Problem-Solving Skills. They should be disciplined, cooperative in team settings and adaptable to work in real job scenarios involving food processing.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [WBSC/AGR/N3201: Apply Safe Working Practices \(Service Assistant Agriculture Machineries\)](#)
2. [WBSC/AGR/N3202: Identify and select various agricultural machineries and describe the uses of each.](#)
3. [WBSC/AGR/N3203: Describe the operations of Tillage implements, demonstrate its working, identify the faults and repair.](#)
4. [WBSC/AGR/N3204: Describe the operations of seeding machinery, its working, identify the faults and repair.](#)
5. [WBSC/AGR/N3205: Recognize different components, different faults, their cause & repair minor faults concern with Harvesting machineries and Threshing machineries.](#)
6. [WBSC/AGR/N3206: Classify IC engines used in agriculture, explain the working of engine system, fuel system, cooling system, lubricating system, inlet & exhaust system with the help of flow charts](#)
7. [WBSC/AGR/N3207: Identify and Describe different component and systems of Tractor and repair minor faults.](#)
8. [WBSC/AGR/N3208: Identify and Recognize different component and systems of Power Tiller and rectify minor faults](#)
9. [WBSC/AGR/N3209: Demonstrate different component and systems of Plant Protection Equipment and rectify minor faults.](#)
10. [WBSC/AGR/N3210: Exhibit different component and systems of Irrigation Equipment and rectify minor faults](#)
11. [WBSC/AGR/N3211: Demonstrate different component and systems of electric motor used in Pump](#)



Qualification Pack

[Set for Agricultural use and rectify minor faults](#)

12. [WBSC/AGR/N3212: Understand principles of pump set, their components, common faults and dismantling different parts](#)

13. [WBSC/AGR/N3213: Calculate rough estimation and costing of repair work for each machinery.](#)

14. [WBSC/AGR/N3214: Able to work in real job situation with special emphasis on basic safety and hazards in this domain.](#)

15. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Agriculture Crop Production
Occupation	Service Assistant (Agriculture Machineries)
Country	India
NSQF Level	3
Credits	20
Aligned to NCO/ISCO/ISIC Code	7233.2800
Minimum Educational Qualification & Experience	10th Class with NA of experience OR 8th grade pass and pursuing continuous schooling in regular school with vocational subject with NA of experience OR 8th Class pass with 2 Years of experience OR 5th Class Pass with 5 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	03/05/2026



Qualification Pack

NSQC Approval Date	03/05/2023
Version	1.0
Reference code on NQR	QG-03-AG-00364-2023-V2-WBSC
NQR Version	2



Qualification Pack

WBSC/AGR/N3201: Apply Safe Working Practices (Service Assistant Agriculture Machineries)

Description

Apply Safe Working Practices

Scope

The scope covers the following :

- Safe Working Practices.
- Recognition of any unsafe situations according to site policy.
- Fire and safety hazards.
- Different types and usage of fire extinguishers as per requirements.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply and maintain Safe Working Practices.
- PC2.** Recognize any unsafe situations according to site policy.
- PC3.** Identify fire and safety and hazards.
- PC4.** Identify different fire extinguishers and use them as per requirements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The procedures to achieve a safe working environment in line with occupational health and safety regulations and requirements according to site policy.
- KU2.** Recognition of any unsafe situations according to site policy, and assess his report accordingly.
- KU3.** Identification and process of taking necessary precautions on fire and safety hazards and report according to site policy and procedures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Recognize any unsafe situations according to site policy, and assess his report accordingly.
- GS2.** Demonstrate Personal Protective Equipment (PPE) like: safety helmet, safety glove, and safety shoe, use the same as per related working environment.
- GS3.** Demonstrate basic first aid & CPR and use them under different circumstances.
- GS4.** Identify different fire extinguishers and use the same as per requirement in a mock drill.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	14	30	-	-
PC1. Apply and maintain Safe Working Practices.	-	-	-	-
PC2. Recognize any unsafe situations according to site policy.	-	-	-	-
PC3. Identify fire and safety and hazards.	-	-	-	-
PC4. Identify different fire extinguishers and use them as per requirements.	-	-	-	-
NOS Total	14	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3201
NOS Name	Apply Safe Working Practices (Service Assistant Agriculture Machineries)
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries), Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3202: Identify and select various agricultural machineries and describe the uses of each.

Description

Identify and select various agricultural machineries and describe the uses of each.

Scope

The scope covers the following :

- Different types of agricultural machineries and illustrate their application.
- Location and identification of the different accessories as per applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Identify different agricultural machineries.

PC2. Illustrate the application of each.

PC3. Locate and identify the different accessories as per applications.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Crop production machinery.

KU2. Tillage implements.

KU3. Seeding machinery.

KU4. Harvesting machinery.

KU5. Threshing machinery.

KU6. Diesel pump set.

KU7. Tractor and power Tiller.

KU8. Plant protection equipment.

KU9. Post-harvest machinery.

KU10. Farm equipment.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify different agricultural machineries.

GS2. Illustrate the application of each.

GS3. Locate and identify the different accessories as per application.

GS4. Identify and operate different Crop production machinery.



Qualification Pack

- GS5.** Use Tractor and power Tiller.
- GS6.** Identify and use different Plant protection equipment.
- GS7.** Identify and use different Post-harvest machinery.
- GS8.** Identify and use different Farm equipment.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	14	30	-	-
PC1. Identify different agricultural machineries.	-	-	-	-
PC2. Illustrate the application of each.	-	-	-	-
PC3. Locate and identify the different accessories as per applications.	-	-	-	-
NOS Total	14	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3202
NOS Name	Identify and select various agricultural machineries and describe the uses of each.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3203: Describe the operations of Tillage implements, demonstrate its working, identify the faults and repair.

Description

Describe the operations of Tillage implements, demonstrate its working, identify the faults and repair.

Scope

The scope covers the following :

- The types and necessity of Tillage.
- Functions and types of implements for primary and secondary tillage operations.
- Constructional parts and working of Mouldboard plough, Disk Plough, Cultivators, Disk Harrow, and Bund former.
- Alignment adjustment, Care taking and safety precautions.
- Common faults and remedies regarding tillage implement.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. State the types and necessity of Tillage.

PC2. Demonstrate functions and types of implements for primary and secondary tillage operations.

PC3. Demonstrate Constructional parts and working of Mouldboard plough, Disk Plough, Cultivators, Disk Harrow, and Bund former.

PC4. Adjust the alignment, take care and safety precautions.

PC5. Demonstrate common faults and remedies regarding tillage implement.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Tillage Implements.

KU2. Tillage definition, types and necessity, Functions and types of implements for primary and secondary tillage operations.

KU3. Constructional parts and working of Mouldboard plough, Disk Plough, Cultivators, Disk Harrow, and Bund former.

KU4. Adjustment, care and safety precautions, common faults and remedies.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. State the types and necessity of Tillage.

GS2. Demonstrate functions and types of implements for primary and secondary tillage operations.



Qualification Pack

- GS3.** Demonstrate Constructional parts and working of Mouldboard plough, Disk Plough, Cultivators, Disk Harrow, and Bund former.
- GS4.** Adjust the alignment, take care and safety precautions.
- GS5.** Demonstrate common faults and remedies regarding tillage implement.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	14	30	-	-
PC1. State the types and necessity of Tillage.	-	-	-	-
PC2. Demonstrate functions and types of implements for primary and secondary tillage operations.	-	-	-	-
PC3. Demonstrate Constructional parts and working of Mouldboard plough, Disk Plough, Cultivators, Disk Harrow, and Bund former.	-	-	-	-
PC4. Adjust the alignment, take care and safety precautions.	-	-	-	-
PC5. Demonstrate common faults and remedies regarding tillage implement.	-	-	-	-
NOS Total	14	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3203
NOS Name	Describe the operations of Tillage implements, demonstrate its working, identify the faults and repair.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3204: Describe the operations of seeding machinery, its working, identify the faults and repair.

Description

Describe the operations of seeding machinery, its working, identify the faults and repair.

Scope

The scope covers the following :

- Different methods of sowing such as drilling, dibbling, planting etc.
- Different implements used for sowing /planting and transplanting.
- Constructions and working principle of seed-drill, planters and seed cum fertilizer drill.
- Common faults and remedies regarding seeding machineries.
- Adjustment of the alignment, taking care and safety precautions regarding seeding machineries.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate different methods of sowing such as drilling, dibbling, planting etc.
- PC2.** Recognize different implements used for sowing /planting and transplanting.
- PC3.** Demonstrate constructions and working principle of seed-drill, planters and seed cum fertilizer drill.
- PC4.** Identify common faults and remedies regarding seeding machineries.
- PC5.** Adjust the alignment, take care and safety precautions regarding seeding machineries.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Methods of sowing- drilling, dibbling, planting, implements for sowing/planting and transplanting.
- KU2.** Constructions and working of seed drill, planters, seed cum fertilizer drill. Adjustment, care and safety precautions, common faults and remedies.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify different parts of seeders, planters and seed cum fertilizer drills.
- GS2.** Adjust furrow openers, replace worn-out / damaged parts of seed drill/ seed cum fertilizer drills.
- GS3.** Calibrate seed drill/ seed cum fertilizer drills.
- GS4.** Familiarize rice transplanters.



Qualification Pack

GS5. Operate seed drill/ seed cum fertilizer drills and rice transplanters.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	14	30	-	-
PC1. Demonstrate different methods of sowing such as drilling, dibbling, planting etc.	-	-	-	-
PC2. Recognize different implements used for sowing /planting and transplanting.	-	-	-	-
PC3. Demonstrate constructions and working principle of seed-drill, planters and seed cum fertilizer drill.	-	-	-	-
PC4. Identify common faults and remedies regarding seeding machineries.	-	-	-	-
PC5. Adjust the alignment, take care and safety precautions regarding seeding machineries.	-	-	-	-
NOS Total	14	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3204
NOS Name	Describe the operations of seeding machinery, its working, identify the faults and repair.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3205: Recognize different components, different faults, their cause & repair minor faults concern with Harvesting machineries and Threshing machineries.

Description

Recognize different components, different faults, their cause & repair minor faults concern with Harvesting machineries and Threshing machineries.

Scope

The scope covers the following :

- Different types of Conventional and mechanized harvesting Tools and Machines such as Sickle, Reaper, Harvester, etc.
- Construction and working principle of Power Tiller operated Vertical Conveyor Reaper.
- Common faults and remedies regarding seeding machineries.
- Rectification and testing of the alignment with safety precautions regarding seeding machineries.
- Different types of Threshing Machinery such as manual and power operated.
- Construction and working principle of hand and paddle threshers, power threshers as well as Combine Harvester.
- Common faults and remedies regarding manually operated and machine operated threshing machineries.
- Adjustment of the alignment, test and safety precautions regarding harvesting and threshing machineries.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate different types of Conventional and mechanized harvesting Tools and Machines such as Sickle, Reaper, Harvester, etc.
- PC2.** Describe construction and working principle of Power Tiller operated Vertical Conveyor Reaper.
- PC3.** Identify common faults and remedies regarding seeding machineries.
- PC4.** Rectify and test the alignment with safety precautions regarding seeding machineries.
- PC5.** Demonstrate different types of Threshing Machinery such as manual and power operated.
- PC6.** Describe construction and working principle of hand and paddle threshers, power threshers as well as Combine Harvester.
- PC7.** Identify common faults and remedies regarding manually operated and machine operated threshing machineries.
- PC8.** Adjust the alignment, test and safety precautions regarding harvesting and threshing machineries.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** Harvesting Machinery - Conventional and mechanized harvesting Tools and Machines, Sickle, Reaper, Harvester.
- KU2.** Construction and working of Power Tiller operated Vertical Conveyor Reaper.
- KU3.** Adjustment, care and safety precautions, common faults and remedies.
- KU4.** Threshing Machinery - Types of threshers - manual and power operated.
- KU5.** Construction and working of hand and paddle threshers, power threshers.
- KU6.** Brief introduction of Combine Harvester.
- KU7.** Adjustment, care and safety precautions, common faults and remedies of manual and power operated threshers.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify different parts of Vertical Conveyor Reaper.
- GS2.** Adjust cutter bar, registration and alignment.
- GS3.** Operate, take care and maintain Vertical Conveyor Reaper.
- GS4.** Identify different parts of power thresher.
- GS5.** Dismantle and assemble power thresher.
- GS6.** Identify mini combine/ combine harvester.
- GS7.** Operate mini combine.
- GS8.** Identify common faults and remedies regarding manually operated and machine operated threshing machineries.
- GS9.** Adjust the alignment, test and safety precautions regarding threshing machineries.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. Demonstrate different types of Conventional and mechanized harvesting Tools and Machines such as Sickle, Reaper, Harvester, etc.	-	-	-	-
PC2. Describe construction and working principle of Power Tiller operated Vertical Conveyor Reaper.	-	-	-	-
PC3. Identify common faults and remedies regarding seeding machineries.	-	-	-	-
PC4. Rectify and test the alignment with safety precautions regarding seeding machineries.	-	-	-	-
PC5. Demonstrate different types of Threshing Machinery such as manual and power operated.	-	-	-	-
PC6. Describe construction and working principle of hand and paddle threshers, power threshers as well as Combine Harvester.	-	-	-	-
PC7. Identify common faults and remedies regarding manually operated and machine operated threshing machineries.	-	-	-	-
PC8. Adjust the alignment, test and safety precautions regarding harvesting and threshing machineries.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3205
NOS Name	Recognize different components, different faults, their cause & repair minor faults concern with Harvesting machineries and Threshing machineries.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3206: Classify IC engines used in agriculture, explain the working of engine system, fuel system, cooling system, lubricating system, inlet & exhaust system with the help of flow charts

Description

Classify IC engines used in agriculture, explain the working of engine system, fuel system, cooling system, lubricating system, inlet & exhaust system with the help of flow charts

Scope

The scope covers the following :

- Different types of IC engine.
- Characteristics and application of two stroke and four stroke engine.
- Different parts and function of an IC engine.
- Components and function of different systems concerned with engine such as engine system, fuel system, cooling system, lubricating system, and inlet & exhaust system with the help of flow chart.
- Dismantlement of the engine, cleanliness of different parts, detection of fault, rectification and refit.
- Different parts such as couplings, pipes, glands, gaskets etc and identification of the defects and repair of the same.
- Alignment test for engine and its adjustment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate different types of IC engine.
- PC2.** Describe the characteristics, application of two stroke and four stroke engine.
- PC3.** Recognize different parts of an IC engine and the function of each.
- PC4.** Describe components and function of different systems concerned with engine such as engine system, fuel system, cooling system, lubricating system, and inlet & exhaust system with the help of flow chart.
- PC5.** Dismantle the engine, clean different parts, detect the fault, rectify and refit.
- PC6.** Identify different parts such as couplings, pipes, glands, gaskets etc.
- PC7.** Identify the defects and perform repairing of the same.
- PC8.** Perform alignment test for engine and its adjustment.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Engine classification - CI and SI Engines, 2S and 4S Engines, working principles of the IC engines and difference between 2S & 4S, CI & SI Engines.
- KU2.** Brief description of Engine components.
- KU3.** Engine Systems- Brief description of engine systems and their necessity.



Qualification Pack

KU4. Fuel system, cooling system, lubrication system, inlet and exhaust system using flow charts.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify IC Engine parts.
- GS2.** Clean and inspect various parts.
- GS3.** Perform Engine overhauling- dismantling, cleaning & adjusting and assembling.
- GS4.** Identify components and function of different systems concerned with engine such as engine system, fuel system, cooling system, lubricating system, and inlet & exhaust system with the help of flow chart.
- GS5.** Dismantle the engine, clean different parts, detect the fault, rectify and refit.
- GS6.** Identify different parts such as couplings, pipes, glands, gaskets etc.
- GS7.** Identify the defects and repair the same.
- GS8.** Perform Alignment test for engine and its adjustment.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. Demonstrate different types of IC engine.	-	-	-	-
PC2. Describe the characteristics, application of two stroke and four stroke engine.	-	-	-	-
PC3. Recognize different parts of an IC engine and the function of each.	-	-	-	-
PC4. Describe components and function of different systems concerned with engine such as engine system, fuel system, cooling system, lubricating system, and inlet & exhaust system with the help of flow chart.	-	-	-	-
PC5. Dismantle the engine, clean different parts, detect the fault, rectify and refit.	-	-	-	-
PC6. Identify different parts such as couplings, pipes, glands, gaskets etc.	-	-	-	-
PC7. Identify the defects and perform repairing of the same.	-	-	-	-
PC8. Perform alignment test for engine and its adjustment.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3206
NOS Name	Classify IC engines used in agriculture, explain the working of engine system, fuel system, cooling system, lubricating system, inlet & exhaust system with the help of flow charts
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3207: Identify and Describe different component and systems of Tractor and repair minor faults.

Description

Identify and Describe different component and systems of Tractor and repair minor faults.

Scope

The scope covers the following :

- Different makes and models of Tractor.
- Different components and systems of Tractor.
- Tractor controls and matching implements.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. State different makes and models of Tractor.

PC2. Describe different components and systems of Tractor.

PC3. State tractor controls and matching implements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Different makes and models of Tractor.

KU2. Different components and systems of Tractor.

KU3. Tractor controls and matching implements.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify different parts and systems of tractor.

GS2. Operate tractor without implements.

GS3. Implement hitching and tractor operation.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. State different makes and models of Tractor.	-	-	-	-
PC2. Describe different components and systems of Tractor.	-	-	-	-
PC3. State tractor controls and matching implements.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3207
NOS Name	Identify and Describe different component and systems of Tractor and repair minor faults.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3208: Identify and Recognize different component and systems of Power Tiller and rectify minor faults

Description

Identify and Recognize different component and systems of Power Tiller and rectify minor faults

Scope

The scope covers the following :

- Different makes and models of Power Tiller.
- Different components and systems of Power Tiller.
- Power tiller controls and matching implements.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. State Different makes and models of Power Tiller.

PC2. Demonstrate different components and systems of Power Tiller.

PC3. State power tiller controls and matching implements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Different makes and models of Power Tiller.

KU2. Different components and systems of Power Tiller.

KU3. Power tiller controls and matching implements.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify different makes and models of Power Tiller.

GS2. Identify different parts and systems of power tiller.

GS3. Operate power tiller with matching implements - roiatiller and MB Plough.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. State Different makes and models of Power Tiller.	-	-	-	-
PC2. Demonstrate different components and systems of Power Tiller.	-	-	-	-
PC3. State power tiller controls and matching implements.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3208
NOS Name	Identify and Recognize different component and systems of Power Tiller and rectify minor faults
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3209: Demonstrate different component and systems of Plant Protection Equipment and rectify minor faults.

Description

Demonstrate different component and systems of Plant Protection Equipment and rectify minor faults.

Scope

The scope covers the following :

- Plant Protection Equipment.
- Use and application of Agricultural Chemicals.
- Types of sprayers & their uses, construction and function.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate Plant Protection Equipment.

PC2. Describe the use and application of Agricultural Chemicals.

PC3. Demonstrate types of sprayers & their uses, construction and function.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Plant Protection Equipment.

KU2. Use and application of Agricultural Chemicals.

KU3. Types of sprayers & their uses, construction and function.

KU4. Types of dusters and their uses, construction and function.

KU5. Safety precautions in handling of chemicals and operational techniques.

KU6. Proper off-season storage.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Operate Plant Protection Equipment.

GS2. Identify different parts of sprayers.

GS3. Dismantle sprayers, replacement of worn out parts

GS4. Calibrate and operate sprayers for specific purposes.

GS5. Identify common faults and corrective measures.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. Demonstrate Plant Protection Equipment.	-	-	-	-
PC2. Describe the use and application of Agricultural Chemicals.	-	-	-	-
PC3. Demonstrate types of sprayers & their uses, construction and function.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3209
NOS Name	Demonstrate different component and systems of Plant Protection Equipment and rectify minor faults.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries), Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3210: Exhibit different component and systems of Irrigation Equipment and rectify minor faults

Description

Exhibit different component and systems of Irrigation Equipment and rectify minor faults

Scope

The scope covers the following :

- Importance and Methods of irrigation.
- Types of pumps, principles of operation, their constructional details and application.
- Estimation of head, discharge and power requirement.
- Importance and utility of Sprinkler irrigation system, drip irrigation system and equipment.
- Pipe fitting techniques and field layouts.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** State Importance of irrigation and methods of Irrigation.
- PC2.** Describe types of pumps, principles of operation, their constructional details and application.
- PC3.** Demonstrate estimation of head, discharge and power requirement.
- PC4.** Demonstrate Sprinkler irrigation system and equipment, importance and utility.
- PC5.** Demonstrate Drip irrigation system and equipment, importance and utility.
- PC6.** Demonstrate pipe fitting techniques and field layouts.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance of irrigation and methods of Irrigation.
- KU2.** Types of pumps, principles of operation, their constructional details and application.
- KU3.** Estimation of head, discharge and power requirement laying of pipes.
- KU4.** Sprinkler irrigation system and equipment, importance and utility.
- KU5.** Drip irrigation system and equipment, importance and utility.
- KU6.** Pipe fitting techniques and field layouts.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate different methods of Irrigation.
- GS2.** Identify different types of pump sets.



Qualification Pack

- GS3.** Dismantle centrifugal pump, reconditioning and assembling.
- GS4.** Install pump, prime mover, fitting of pumps, valves, pulleys.
- GS5.** Identify different components of sprinkler and drip irrigation systems.
- GS6.** Dismantle and assemble nozzles, drips and other associated components of an irrigation system.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. State Importance of irrigation and methods of Irrigation.	-	-	-	-
PC2. Describe types of pumps, principles of operation, their constructional details and application.	-	-	-	-
PC3. Demonstrate estimation of head, discharge and power requirement.	-	-	-	-
PC4. Demonstrate Sprinkler irrigation system and equipment, importance and utility.	-	-	-	-
PC5. Demonstrate Drip irrigation system and equipment, importance and utility.	-	-	-	-
PC6. Demonstrate pipe fitting techniques and field layouts.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3210
NOS Name	Exhibit different component and systems of Irrigation Equipment and rectify minor faults
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3211: Demonstrate different component and systems of electric motor used in Pump Set for Agricultural use and rectify minor faults

Description

Demonstrate different component and systems of electric motor used in Pump Set for Agricultural use and rectify minor faults

Scope

The scope covers the following :

- Types of electric motors used in pump sets, tube wells, threshers etc.
- Selection method, care, maintenance, installation, circuit checking, starter fitting and setting of electric motors.
- Electric motor trouble shooting, periodic servicing and off-season storage.
- Safety and precautions in handling electrical appliances & motors.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate types of electric motors used in pump sets, tube wells, threshers etc.

PC2. Identify the selection method of electric motors.

PC3. Demonstrate about care, maintenance and installation of electric motors.

PC4. Demonstrate electric motor trouble shooting, periodic servicing and off-season storage.

PC5. Install electric motor, check the circuit, starter fitting and setting.

PC6. Demonstrate safety, precautions in handling electrical appliances & motors.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Types of electric motors used in pump sets, tube wells, threshers etc. Selection of electric motors.

KU2. Care, maintenance and installation of electric motors.

KU3. Electric motor trouble shooting, periodic servicing and off-season storage Installation of electric motor, checking of circuit, starter fitting and setting.

KU4. Safety, precautions in handling electrical appliances & motors.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify types of electric motors used in pump sets, tube wells, threshers etc.



Qualification Pack

- GS2.** Identify the selection method of electric motors.
- GS3.** Take care, maintain and install electric motors.
- GS4.** Demonstrate electric motor trouble shooting, periodic servicing and off-season storage.
- GS5.** Install electric motor, check the circuit, starter fitting and setting.
- GS6.** Maintain safety, precautions in handling electrical appliances & motors.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. Demonstrate types of electric motors used in pump sets, tube wells, threshers etc.	-	-	-	-
PC2. Identify the selection method of electric motors.	-	-	-	-
PC3. Demonstrate about care, maintenance and installation of electric motors.	-	-	-	-
PC4. Demonstrate electric motor trouble shooting, periodic servicing and off-season storage.	-	-	-	-
PC5. Install electric motor, check the circuit, starter fitting and setting.	-	-	-	-
PC6. Demonstrate safety, precautions in handling electrical appliances & motors.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3211
NOS Name	Demonstrate different component and systems of electric motor used in Pump Set for Agricultural use and rectify minor faults
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3212: Understand principles of pump set, their components, common faults and dismantling different parts

Description

Understand principles of pump set, their components, common faults and dismantling different parts

Scope

The scope covers the following :

- General ideas on pumps along with common faults concerned with pumps.
- Dismantlement and reassembling of the parts to make a complete pump set.
- Alignment test and its adjustment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate general ideas on pumps along with common faults concerned with pumps.

PC2. Able to dismantle and reassemble the parts to make a complete pump set.

PC3. Able to perform Alignment test and its adjustment.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Diesel pump set used in agriculture.

KU2. Different types of diesel engines-idea of 2-stroke & 4-stroke engines- single cylinder & multi-cylinder engines.

KU3. Different parts of a diesel engine, identification by physical verification & function of each part.

KU4. Different parts of pump, Identification by physical observation & function of each part.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Practice fitting, use file, chisel, hammer, vice, measuring tools etc.

GS2. Practice gas & arc welding on simple job & take precautions during the work.

GS3. Identify & use different repairing jobs tools, measuring tools, special tools e.g., puller, piston ring expander, piston, ring squeezer, groove cleaner, valve lifter, torque wrench, grease gun etc. and maintain & service them.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	50	-	-
PC1. Demonstrate general ideas on pumps along with common faults concerned with pumps.	-	-	-	-
PC2. Able to dismantle and reassemble the parts to make a complete pump set.	-	-	-	-
PC3. Able to perform Alignment test and its adjustment.	-	-	-	-
NOS Total	10	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3212
NOS Name	Understand principles of pump set, their components, common faults and dismantling different parts
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3213: Calculate rough estimation and costing of repair work for each machinery.

Description

Calculate rough estimation and costing of repair work for each machinery.

Scope

The scope covers the following :

- Costing for different parts, their costs, brand names of parts, costs etc.,
- Report making on Market survey.
- Estimation for a repair work-overhauling of an engine etc.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Calculate the costing for different parts, their costs, brand names of parts, costs etc.

PC2. Make the report on Market survey.

PC3. Prepare of an estimate for a repair work-overhauling of an engine etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Fixed cost: initial cost, salvage value, useful life, depreciation, interest, taxes and insurance, housing.

KU2. Variable cost: repair and maintenance, fuel and oil consumption, wages.

KU3. Theoretical and actual field capacities, field efficiency.

KU4. The calculation of the costing for different parts, their costs, brand names of parts, costs etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Calculate cost per hour/per hectare of tractors, power tillers, engine, and various agricultural implements and machines.

GS2. Lay out a store, fabrication racks; familiarization with fire extinguishers, visit to a store.

GS3. Calculate the costing for different parts, their costs, brand names of parts, costs etc.,

GS4. Make the report on Market survey.

GS5. Prepare an estimate for a repair work- overhauling of an engine etc.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	14	30	-	-
PC1. Calculate the costing for different parts, their costs, brand names of parts, costs etc.	-	-	-	-
PC2. Make the report on Market survey.	-	-	-	-
PC3. Prepare of an estimate for a repair work-overhauling of an engine etc.	-	-	-	-
NOS Total	14	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3213
NOS Name	Calculate rough estimation and costing of repair work for each machinery.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3214: Able to work in real job situation with special emphasis on basic safety and hazards in this domain.

Description

Able to work in real job situation with special emphasis on basic safety and hazards in this domain.

Scope

The scope covers the following :

- Work in the real job situation.
- Report preparation on the competency developed.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Work in the real job situation with special emphasis on basic safety and hazards in this domain.
- PC2.** Prepare report for this component of training of the course to ensure whether competency has been developed.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Complete Production jobs as per drawing such as furniture items.
- GS2.** Prepare utility goods for domestic use by steel wire, preparation of different models with 5mm. M.S. Rods or wire for common structural items - grills, Gratings, Gates etc.
- GS3.** Prepare report for this component of training of the course to ensure whether competency has been developed to work in the real job situation with special emphasis on basic safety and hazards in this domain.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	250	-	-
PC1. Work in the real job situation with special emphasis on basic safety and hazards in this domain.	-	-	-	-
PC2. Prepare report for this component of training of the course to ensure whether competency has been developed.	-	-	-	-
NOS Total	-	250	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3214
NOS Name	Able to work in real job situation with special emphasis on basic safety and hazards in this domain.
Sector	Agriculture
Sub-Sector	
Occupation	Service Assistant (Agriculture Machineries)
NSQF Level	3
Credits	5
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment will be based on the concept of Independent Assessors empaneled with West Bengal State Council of Technical & Vocational Education & Skill Development (WBSCT&VE&SD), identified, selected, trained and certified on Assessment techniques. These Assessors would be aligned to assess as per the laid down criteria.

WBSCT&VE&SD would conduct assessment only at the training centers or designated testing centers authorized by WBSCT&VE&SD. Ideally, the assessment will be a continuous process comprising of two distinct steps:

- Continuous assessment by Trainers
- Term end /Final Assessment by WBSCT&VE&SD

Each National Occupational Standard (NOS) in the respective QPs will be assigned weightage. Each Performance Criteria in the NOS will be assigned marks for theory and/or practical based on relative importance and criticality of function. This will facilitate preparation of question bank / paper sets for each of the QPs. Each of these papers sets/question banks created by subject matter experts through WBSCT&VE&SD, especially with regard to the practical test and the defined tolerances, finish, accuracy



Qualification Pack

etc. The following tools are proposed to be used for final assessment:

i. Written Test: This will comprise of (i) True/False Statements and/or (ii) Multiple Choice Questions and/or (iii) Matching Type Questions. Online system for this will be preferred.

ii. Practical Test: This will comprise a test job to be prepared as per project briefing following appropriate working steps, using necessary tools, equipment and instruments. Through observation it will be possible to ascertain candidate's aptitude, attention to details, quality consciousness etc.

iii. Structured Viva-voce: This tool will be used to assess the conceptual understanding and the behavioral aspects as regards the job role and the specific task at hand.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/AGR/N3201. Apply Safe Working Practices (Service Assistant Agriculture Machineries)	14	30	0	0	44	4
WBSC/AGR/N3202. Identify and select various agricultural machineries and describe the uses of each.	14	30	0	0	44	4
WBSC/AGR/N3203. Describe the operations of Tillage implements, demonstrate its working, identify the faults and repair.	14	30	0	0	44	5
WBSC/AGR/N3204. Describe the operations of seeding machinery, its working, identify the faults and repair.	14	30	0	0	44	5



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/AGR/N3205. Recognize different components, different faults, their cause & repair minor faults concern with Harvesting machineries and Threshing machineries.	10	50	0	0	60	6
WBSC/AGR/N3206. Classify IC engines used in agriculture, explain the working of engine system, fuel system, cooling system, lubricating system, inlet & exhaust system with the help of flow charts	10	50	0	0	60	6
WBSC/AGR/N3207. Identify and Describe different component and systems of Tractor and repair minor faults.	10	50	0	0	60	6
WBSC/AGR/N3208. Identify and Recognize different component and systems of Power Tiller and rectify minor faults	10	50	0	0	60	6
WBSC/AGR/N3209. Demonstrate different component and systems of Plant Protection Equipment and rectify minor faults.	10	50	0	0	60	6
WBSC/AGR/N3210. Exhibit different component and systems of Irrigation Equipment and rectify minor faults	10	50	0	0	60	6
WBSC/AGR/N3211. Demonstrate different component and systems of electric motor used in Pump Set for Agricultural use and rectify minor faults	10	50	0	0	60	6
WBSC/AGR/N3212. Understand principles of pump set, their components, common faults and dismantling different parts	10	50	0	0	60	6



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/AGR/N3213.Calculate rough estimation and costing of repair work for each machinery.	14	30	0	0	44	4
WBSC/AGR/N3214.Able to work in real job situation with special emphasis on basic safety and hazards in this domain.	0	250	0	0	250	25
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	170	830	-	-	1000	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NQR	National Qualification Register
OJT	On the Job Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.



Qualification Pack

National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an N
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding KU	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills GS	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today-s world. These skills are typically needed in any work environment in today-s world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.